

Back Pain - Non Invasive Treatments - Medical Clinical Policy Bulletins

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Policy

Scope of Policy

This Clinical Policy Bulletin addresses non-invasive treatments for back pain.

1. Medical Necessity

1. Back School

Aetna considers back school medically necessary for the treatment of persons with chronic or recurrent back pain, when such a program is prescribed by the member's doctor and the program is conducted by a physical therapist or other appropriate recognized healthcare professional. See [CPB 0325 - Physical Therapy](#).

2. Isokinetic Devices

Isokinetic devices (e.g., Biodex, Cybex, and Kin-Com) and other exercise and testing machines (e.g., Isostation B-2000 and MedX) are considered acceptable alternatives for provision of medically necessary exercise in physical therapy. In addition to use in muscle testing, the MedX and other machines have also been used for administering exercise therapy. These devices can be used as exercise machines for administering physical therapy. However, these particular brands of exercise devices have not been proven to be superior to standard brands of exercise equipment (e.g., Nautilus, etc.) when used for administering physical therapy.

2. Experimental, Investigational, or Unproven

The following interventions are considered experimental, investigational, or unproven because the effectiveness of these approaches has not been established:

1. Quantitative Muscle Testing Devices

The use of quantitative muscle testing devices (not an all-inclusive list) when used for muscle testing because there is insufficient evidence that use of these devices improves the assessment of muscle strength over standard manual strength testing such that clinical outcomes are improved:

1. MedX Lumbar and Cervical Extension Devices
2. Isostation B-200 Lumbar Dynamometer
3. Kin-Com Physical Therapy Isokinetic Equipment
4. Cybex Back System
5. Biodex System 3
6. JTECH Tracker Freedom Wireless Muscle Testing.

Note: No additional reimbursement is provided for performing manual muscle testing using hand-held dynamometers (not an all-inclusive list):

1. Lafayette Manual Muscle Test
2. Nicholas Manual Muscle Tester
3. Hoggan Dynamometer.

The use of the hand-held dynamometer is considered integral to the manual muscle testing and is not separately reimbursed.

2. Other Interventions for the Treatment of Back Pain

1. Auricular acupressure
2. Combined capacitive resistive monopolar radiofrequency and proprioceptive neuromuscular facilitation
3. Cupping therapy
4. Dr. Ho's 2-in-1 Decompression Belt
5. Gabapentinoids (e.g., gabapentin and pregabalin) (excluding fibromyalgia indication)
6. High-frequency impulse therapy
7. Intermittent vertical lumbar traction
8. Khan Kinetic Treatment
9. Kinesiotaping
10. Matrix rhythm therapy
11. Orthotrac pneumatic vest
12. Spinal adjusting instruments (see [CPB 0107 - Chiropractic Services](#)).
13. Sustained acoustic medicine.

3. Policy Limitations and Exclusions

Back school for occupational purposes may be excluded from coverage. See [CPB 0250 - Occupational Therapy](#) and [CPB 0198 - Work Hardening Programs](#). Please check benefit plan descriptions for details.

4. Related Policies

- [CPB 0011 - Electrical Stimulation for Pain](#)
- [CPB 0016 - Back Pain - Invasive Procedures](#)
- [CPB 0107 - Chiropractic Services](#)
- [CPB 0132 - Biofeedback](#)
- [CPB 0135 - Acupuncture](#)
- [CPB 0180 - Vertebral Axial Decompression](#)
- [CPB 0198 - Work Hardening Programs](#)

- [CPB 0204 - Manipulation Under General Anesthesia](#)
- [CPB 0207 - Prolotherapy and Sclerotherapy](#)
- [CPB 0250 - Occupational Therapy](#)
- [CPB 0325 - Physical Therapy](#)
- [CPB 0569 - Lumbar Traction Devices](#)

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
CPT codes covered if criteria are met:	
97110	Therapeutic procedure, one or more areas, each 15 minutes; therapeutic exercises to develop strength and endurance, range of motion and flexibility
97140	Manual therapy techniques (e.g., mobilization/manipulation, manual lymphatic drainage, manual traction), one or more regions, each 15 minutes
97530	Therapeutic activities, direct (one-on-one) patient contact (use of dynamic activities to improve functional performance), each 15 minutes [not covered for Khan Kinetic Treatment]
CPT codes not covered for indications listed in the CPB:	
<i>Hand held dynamometer, auricular acupressure, cupping therapy, Combined capacitive resistive monopolar radiofrequency and proprioceptive neuromuscular facilitation, Matrix rhythm therapy - no specific code</i>	
97012	Application of a modality to 1 or more areas; traction, mechanical
97545	Work hardening/conditioning; initial 2 hours
+97546	each additional hour (List separately in addition to code for primary procedure)
97750	Physical performance test or measurement (e.g., musculoskeletal, functional capacity), with written report, each 15 minutes
Other CPT codes related to the CPB:	
96041	Medical genetics and genetic counseling services, each 30 minutes of total time provided by the genetic counselor on the date of the encounter
HCPCS codes covered if selection criteria are met:	
S9117	Back school, per visit
HCPCS codes not covered for indications listed in the CPB:	

Code	Code Description
<i>Gabapentinoids (e.g., gabapentin and pregabalin), Dr. Ho's 2-in-1 Decompression Belt, high-frequency impulse therapy, kinesiotaping - no specific code:</i>	
E0830	Ambulatory traction device, all types, each
K1004	Low frequency ultrasonic diathermy treatment device for home use, includes all components and accessories (sustained acoustic medicine)
ICD-10 codes covered if selection criteria are met:	
M54.00 - M54.9	Dorsalgia

Background

Quantitative Muscle Testing Devices

Muscle strength testing may be used to determine bilateral differences in strength or other differences in patient resistance. These differences may be characterized by the experienced examiner based on various technologies, i.e., manual, mechanized and computerized muscle testing. These changes may be a result of alterations of function at various levels of the neuromuscular system and/or any other system related to the patient. Computerized muscle testing has been used in clinical research to quantify muscle strength and enables the investigator to produce comparison reports, quantifying patient response to rehabilitation and therapy. However, manual muscle testing is sufficiently reliable for clinical practice. There is insufficient peer-reviewed published scientific evidence that computerized muscle testing leads to better patient outcomes.

The MedX lumbar/cervical extension machine has been proposed for use for isometric and isokinetic muscle testing and rehabilitation in persons with low back pain (LBP) and/or neck pain. The MedX lumbar/cervical extension device has not been adequately validated as a test of isometric and isokinetic muscle strength in persons with back or neck pain. In addition, the MedX machine has not been shown to be superior to any other particular brand of exercise equipment when used for administering physical therapy. A technology assessment of the MedX Lumbar Extension Machine for the treatment of LBP by the Washington State Department of Labor and Industries (2003) concluded: "The evidence suggests that MedX may help to increase lumbar muscle strength. However, studies do not clearly show MedX's efficacy over other exercise programs."

Although there is limited evidence that when used as a training device, the MedX system can help to increase the strength of the lumbar as well as the cervical extensors, it has not been proven that the MedX machines are more effective than standard exercise equipment in functional strengthening. Further investigation, especially controlled studies with pain patients is needed to demonstrate the accuracy of differentiation between normal subjects and patients, especially patients with non-spinal cord injuries of the cervical spine. Additionally, more research is needed to define the contribution of this equipment to patient management, especially in relation to the significant outcomes of psychological distress, changes in daily activities, and ability to return to work in such patients.

The Isostation B-200 lumbar dynamometry equipment has been suggested for use for the evaluation and rehabilitation of persons with LBP. Available evidence fails to establish the clinical effectiveness and significance of the use of the Isostation B-200 lumbar dynamometer for isometric and isokinetic muscle testing (spinal motion and trunk function) and rehabilitation in patients with LBP. More research is needed to establish the ability of this technology to discriminate between normal subjects and patients, to establish test-retest reliability, and to define its contribution to and role in patient management. Additionally, further research is needed to evaluate the relationship

between dynamometric technology, psychological tests and behavior assessments.

The Cybex back system has been proposed for use for evaluation and rehabilitation of persons with LBP. In addition, the Cybex back system has not been proven to be superior to any other particular brand of exercise equipment for administering physical therapy.

It has not been proven that the Cybex system is more effective than standard exercise equipment in functional strengthening. More research is needed to increase confidence in interpretation of abnormal range of motion and strength data, to define rehabilitation goals, and more importantly to define the contribution of this equipment to the management of persons with LBP, especially in relation to health outcomes.

Other brands of isokinetic devices used for quantification of muscle strength include the Kin-Com Isokinetic Muscle Testing Device and the Biodex Muscle Testing Device.

In a cross-sectional study, Gruther and colleagues (2009) examined the accuracy and long-term reliability of dynamometric trunk muscle strength and endurance tests in patients with chronic LBP. A total of 32 patients with chronic LBP, 19 healthy controls and 15 patients with chronic headache matched for age, sex and body mass index were included in the study. Both patient groups and healthy controls performed isokinetic and isometric trunk extensor and flexor tests on a Biodex 2000 dynamometer. The Biering-Sørensen test served to examine back muscle endurance. Borg-Category-Ratio-Scales CR-10 rated participants' body experience immediately before and after the testing. Patients with chronic LBP repeated measurements after 3 weeks. Among dynamometric tests, isokinetic measurements revealed the best area under the curve (AUC = 0.89) for the discrimination between patients with chronic LBP and healthy controls. Reliability testing revealed highly significant learning effects for isometric trunk flexion and isokinetic measurements. The Biering-Sørensen test demonstrated excellent accuracy (AUC = 0.93) and no learning effects. Borg-category-ratio-scale ratings were not associated with the observed changes. The authors concluded that in patients with chronic LBP, dynamometric trunk muscle measures are limited to muscle function assessment purposes. Monitoring treatment outcome in these patients with these measures appears to be problematic because of learning effects. Based on these findings, the authors recommended the Biering-Sørensen test for management of chronic LBP rehabilitation.

Hand-Held Dynamometers

According to the manufacturer, the Lafayette Manual Muscle Testing (MMT) System is an ergonomic hand-held device for objectively quantifying muscle strength. The test is performed with the clinician applying force to the limb of a patient. The objective of the test is for the clinician to overcome or "break" the patient's resistance. The MMT records the peak force and the time required to achieve the "break" providing reliable, accurate, and stable muscle strength readings that conform to most manual muscle testing protocols. The manufacturer states that the Lafayette MMT also has customizable options for data storage, preset test times, and force thresholds.

Published data on the Lafayette MMT includes small studies of its use in a research setting (Tsimaras et al, 2004; Klygite et al, 2003). There are no data on clinical outcomes with the use of the device. Other brands of hand-held dynamometers include the NIcholas Manual Muscle Tester and the Hoggan microFET.

Khan Kinetic Treatment

The Khan Kinetic Treatment (KKT), manufactured by Datrend Systems Inc (Richmond, British Columbia, Canada), is a medical device for the treatment of spine-related abnormalities causing pain. According to the manufacturer, the KKT uses high-frequency small-amplitude sinusoidal waves to vibrate the vertebrae and repeatedly activate associated neuromuscular structures, which evoke multiple mechanisms of pain relief. There is also a small unblinded randomized trial without placebo control, which found that, compared with a control group, the treatment group lowered both their self-recorded neck pain scores ($p = 0.012$) and pain medication dose ($p = 0.048$), although current functional assessment questionnaires (range of motion, overall activity, and recreation/work activities) did not detect changes ($p = 0.233$, 0.311 , and 0.472 , respectively) (Desmoulin et al, 2007). Limitations of this study included a lack of blinding and lack of placebo control. Other published literature on KKT spine treatment consists of a study of the effect of KKT treatment in an animal model (Desmoulin et al, 2010).

JTECH Tracker System

The JTECH Tracker Freedom Wireless Muscle Testing is designed for testing and documentation of strength loss due to injury or disease. However, there is a lack of evidence regarding the effectiveness of the JTECH Tracker muscle testing system.

Orthotrac Pneumatic Vest

The Orthotrac is an inflatable pneumatic vest that has been used to relieve back pain from a variety of causes (e.g., herniated disc, spinal stenosis, facet syndrome, spondylolisthesis, etc). There is insufficient peer-reviewed published clinical evidence of the effectiveness of the Orthotrac pneumatic vest in relieving back pain.

Auricular Acupressure

In a systematic review and meta-analysis, Yang and associates (2017) examined the effects of auricular acupressure (AA) on pain and disability for patients with chronic LBP. These investigators carried out a search of randomized controlled trials (RCTs) in 4 English medical electronic databases and 3 Chinese databases; 2 reviewers independently retrieved related studies, assessed the methodological quality, and extracted data with a standardized data form. Meta-analyses were performed using all time-points meta-analysis (ATM). A total of 7 trials met the inclusion criteria, of which 4 had the low risk of bias. The findings of this study showed that, for the immediate effect, AA had large, significant effects in improving pain within 12 weeks. As for the follow-up effect, the pooled estimates also showed promising effect at 4-week follow-up after 4-week intervention (standardized mean difference [SMD] = -1.13, 95 % CI: -1.70 to -0.56), $p < 0.001$. However, for the disability level, the therapeutic effect of AA was not significant (MD = -1.99, 95 % CI: -4.93 to 0.95), $p = 0.18$. No serious adverse effects were recorded. The authors concluded that the encouraging evidence of this study indicated that AA can be provided to patients with chronic LBP. However, they stated that there is a pressing need for further rigorously designed large-scale RCTs on the effects of AA in patients with chronic LBP.

The authors stated that this study had several drawbacks:

Moreover, only published studies were included in this study, leaving the unpublished negative results out of consideration may lead to the less powerful results.

Cupping Therapy

In a meta-analysis, Wang and colleagues (2017) evaluated the safety and effectiveness of cupping therapy for the patients with LBP. PubMed, Cochrane Library databases, and Embase database were electronically researched; RCTs reporting the cupping for the patients with LBP were included. The meta-analysis was conducted using Review Manager software (version 5.3, Nordic Cochrane Centre). The primary outcome was VAS scores; the secondary outcomes included ODI scores, McGill Present Pain Intensity (MPPI) scores and complications. A total of 6 RCTs were included in this synthesized analysis. The results showed that cupping therapy was superior to the control management with respect to VAS scores (standardized mean difference [SMD]: -0.73, [95 % CI: -1.42 to -0.04]; $p = 0.04$), and ODI scores (SMD: -3.64, [95 % CI: -5.85 to -1.42]; $p = 0.001$). There was no statistical significant difference as regard to MPPI scores. No serious adverse event (AE) was reported in the included studies. The authors concluded that cupping therapy could significantly decrease the VAS scores and ODI scores for patients with LBP compared to the control management. Moreover, they stated that high heterogeneity and risk of bias existing in studies limited the authenticity of the findings.

Gabapentinoids

In a systematic review and meta-analysis, Shanthanna and colleagues (2017) evaluated the safety and effectiveness of gabapentinoids in adult chronic LBP (CLBP) patients. Electronic databases of Medline, Embase, and Cochrane were searched from their inception until December 20, 2016. These researchers included RCTs reporting the use of gabapentinoids for the treatment of CLBP of greater than 3 months duration, in adult patients. Study selection and data extraction was performed independently by paired reviewers. Outcomes were guided by Initiative on Methods, Measurement and Pain

Assessment in Clinical Trials guidelines, with pain relief and safety as the primary outcomes. Meta-analyses were performed for outcomes reported in 3 or more studies. Outcomes were reported as mean differences (MDs) or RRs with their corresponding 95 % CIs, and I² in percentage representing the percentage variability in effect estimates that could be explained by heterogeneity. GRADE (Grading of Recommendations Assessment, Development, and Evaluation) was used to assess the quality of evidence. Out of 1,385 citations, 8 studies were included. Based on the interventions and comparators, studies were analyzed in 3 different groups. Gabapentin (GB) compared with placebo (3 studies, n = 185) showed minimal improvement of pain (MD = 0.22 units, 95 % CI: -0.5 to 0.07, I² = 0 %; GRADE: very low); 3 studies compared pregabalin (PG) with other types of analgesic medication (n = 332) and showed greater improvement in the other analgesic group (MD = 0.42 units, 95 % CI: 0.20 to 0.64, I² = 0; GRADE: very low). Studies using PG as an adjuvant (n = 423) were not pooled due to heterogeneity, but the largest of them showed no benefit of adding PG to tapentadol. There were no deaths or hospitalizations reported. Compared with placebo, the following AEs were more commonly reported with GB: dizziness (RR = 1.99, 95 % CI: 1.17 to 3.37, I² = 49); fatigue (RR = 1.85, 95 % CI: 1.12 to 3.05, I² = 0); difficulties with mentation (RR = 3.34, 95 % CI: 1.54 to 7.25, I² = 0); and visual disturbances (RR = 5.72, 95 % CI: 1.94 to 16.91, I² = 0). The number needed to harm with 95 % CI for dizziness, fatigue, difficulties with mentation, and visual disturbances were 7 (4 to 30), 8 (4 to 44), 6 (4 to 15), and 6 (4 to 13) respectively. The GRADE evidence quality was noted to be very low for dizziness and fatigue, low for difficulties with mentation, and moderate for visual disturbances. Functional and emotional improvements were reported by few studies and showed no significant improvements. The authors concluded that existing evidence on the use of gabapentinoids in CLBP is limited and demonstrated significant risk of adverse effects without any demonstrated benefit. They stated that given the lack of effectiveness, risks, and costs associated, the use of gabapentinoids for CLBP merits caution. Moreover, they stated that there is need for large high-quality clinical trials to more definitively inform this issue.

Fibromyalgia (FM) includes symptoms of widespread musculoskeletal pain. Furthermore, FM may complicate regional pain disorders such as chronic low back pain. The American College of Rheumatology (ACR) guideline-approved fibromyalgia medications include gabapentinoids, such as pregabalin and gabapentin (Goldenberg, 2017).

Decompression Belt

Cannon and colleagues (2016) evaluated the ability of a pneumatic decompression belt to restore spinal height lost following an acute bout of exercise that induced compression.

This study implemented a test-retest repeated measures design in which 12 participants (2 women and 10 men) aged 21.5 ± 1.0 years; height, 179.0 ± 7.70 cm; weight, 84.0 ± 11.5 kg; were recruited from a university population and acted as their own control. All participants were healthy with no previous history of disabling back pain, and were frequent weight-trainers. A stadiometer was used to measure spinal height at baseline, then following an acute bout of exercise and then again following the intervention (use of a pneumatic decompression belt for 20 mins) or control (lying supine for 20 mins). A 2-way repeated measures ANOVA was performed on the change in spinal height in order to evaluate differences between measurement phases and intervention conditions. The use of the decompression belt increased spinal height gain (4.3 ± 3.0 mm) significantly more than the control condition (1.8 ± 1.2 mm) following an acute bout of weight-lifting exercises known to elicit high compressive loads on the lumbar spine. The authors concluded that the pneumatic decompression belt restored spinal height faster than a non-belt wearing condition in young healthy asymptomatic participants. This was a small study (n = 12); and it did not include patients with back pain.

Furthermore, an UpToDate review on "Subacute and chronic low back pain: Nonpharmacologic and pharmacologic treatment" (Chou, 2018) does not mention decompression or pneumatic belts as a therapeutic option.

Mindfulness-Based Cognitive Therapy

Pei and colleagues (2021) noted that chronic pain is a significant public health problem with emotional and disabling factors, which may not completely respond to current medical treatments such as opioids. In a systematic review and meta-analysis, these researchers examined the safety and effectiveness of mindfulness-based cognitive therapy (MBCT) for patients with chronic pain. They carried out database searches of PubMed, Medline, Embase, the Cochrane Library, PsycINFO, Web of Science, Scopus and CINAHL up to October

15, 2019. Included studies evaluated with the Cochrane risk-of-bias tool. A total of 8 RCTs (433 patients), including chronic LBP, fibromyalgia, migraine, rheumatoid arthritis (RA) and mix etiology. MBCT intervention showed a short-term improvement on depression mood [SMD -0.72; 95 % confidence interval (CI): -1.22 to -0.22, $p = 0.005$] compared with usual care and was associated with short-term improvement in mindfulness compared with non-MBCT [SMD 0.51; 95 % CI: 0.01 to 1.01, $p = 0.04$]. Between-group differences in pain intensity, pain inference and pain acceptance were not significant at short- or long-term follow-up. Compared to active treatments, MBCT intervention not found significant differences in either short- or long-term outcomes. The authors concluded that MBCT showed short-term effectiveness on depressed mood and mindfulness of chronic pain patients; however, longer follow-ups, large sample and rigorous RCTs that can best understand remaining uncertainties are needed.

High-Frequency Impulse Therapy

In a randomized-controlled, multi-center, pilot study, Amirdelfan and colleagues (2021) examined the impact of high-frequency impulse therapy (HFIT) on pain and function among patients with CLBP. These researchers carried out a trial of HFIT system versus sham across 5 orthopedic and pain center sites in California. A total of 36 patients seeking treatment for CLBP were randomized. Primary outcome was function measured by the Six-Minute Walk Test (6MWT). Secondary outcomes were function (Timed Up and Go [TUG] and Oswestry Disability Index [ODI]), pain (Numerical Rating Scale [NRS]), quality of life (QOL; Patient Global Impression of Change [PGIC]), and device use. Patients were evaluated at baseline and every week for 4 weeks of follow-up. Mann-Whitney U-test was used to analyze changes in each outcome. Repeated measures ANOVA was used to assess the effect of treatment over time. The average age of subjects was 53.9 ± 15.7 (mean $\pm$ SD) years, with 12.1 ± 8.8 years of CLBP. Patients who received an HFIT device had a significantly higher 6MWT score at weeks 2 (Cohen's d ; 95 % CI: 0.33 (0.02 to 0.61)), 3 [0.32 (0.01 to 0.59)] and 4 [0.31 (0.01 to 0.60)], respectively, as compared to their baseline scores ($p < 0.05$). Patients in the treatment group had significantly lower TUG scores at week 3 [0.30 (0.04 to 0.57)] and significantly lower NRS scores at weeks 2 [0.34 (0.02 to 0.58)] and 4 [0.41 (0.10 to 0.67)] ($p < 0.05$). The authors concluded that a larger-scale RCT can build on the findings of this study to examine if HFIT is effective in reducing pain and improving function in patients with CLBP. These researchers stated that the findings of this study showed encouraging evidence of functional improvement and reduction in pain in subjects who used HFIT. Moreover, these investigators stated that although the results presented support HFIT therapy as a viable treatment for CLBP, the data were from a pilot study with short-term follow-up results at 4 weeks; thus, a RCT is needed for more generalizable results. While this pilot study adhered to strict inclusion and exclusion criteria, enrollment and assessment numbers will be documented in future studies. This study also allowed for full patient control of their use of HFIT; therefore, it was possible that users would expect a sensation from the HFIT device, even though they were informed they should not.

Kinesiotaping

In a meta-analysis, Lin and colleagues (2020) examined the results of RCTs on the effectiveness of Kinesiotaping (KT) for chronic non-specific LBP (CNLBP) and disability. These investigators searched Medline, Cochrane Library, Google Scholar, Web of Science, and Embase from inception to September 1, 2018. Studies were included in the review if they met the following criteria: RCTs published in English; patients (greater than 18 years of age) diagnosed with CNLBP (pain duration of more than 12 weeks), with or without leg pain; KT as a single treatment or as a part of other forms of physical therapy (PT); outcomes measured included pain intensity and disability. Three independent investigators completed data extraction. Methodological quality was appraised using the Cochrane tool for assessing the risk of bias. The GRADE guidelines were used to evaluate the confidence of the effect estimates. A total of 11 RCT studies involving 785 patients were retained for the meta-analysis; SMDs with 95 % CIs were calculated using a random-effects model. Compared with the control group, the pooled SMD of pain intensity was significantly reduced (SMD = -0.73; 95 % CI: -1.12 to -0.35; GRADE: low) and disability was improved (SMD = -0.51; 95 % CI: -0.85 to -0.17; GRADE: low) in the KT group.

Subgroup analyses showed that, compared with the control, the I strip of KT significantly reduced pain (SMD = -0.48; GRADE: low) but not disability (SMD = -0.26; GRADE: low). Compared with sham/placebo tape, KT provided significant pain reduction (SMD = -0.84; GRADE: low) and disability improvement (SMD = -0.56; GRADE: low).

Moreover, compared with the no-tape group, the KT group also showed pain reduction (SMD

= -0.74; GRADE: low) and disability improvement (SMD = -0.65; GRADE: low).

The authors concluded that there is low-quality evidence that KT had a beneficial role in pain reduction and disability improvement for patients with CNLBP. Moreover, these researchers stated that more high-quality studies are needed to confirm the effects of KT on CNLBP. These investigators noted that drawbacks of this meta-analysis included a lack of homogeneity, different methodologies and treatment duration of KT application, and relatively small sample sizes.

Cheatham and associates (2021) stated that the existing body of kinesiology tape (KT) research reveals inconsistent results that challenges the effectiveness of the intervention. Understanding professional beliefs and KT clinical application might provide insight for future research and development of evidence-based guidelines. In a cross-sectional survey study, these researchers documented the beliefs and clinical application methods of KT among healthcare professionals in the U.S. A 30-question online survey was e-mailed to members of the National Athletic Trainers Association, Academy of Orthopedic Physical Therapy, and American Academy of Sports Physical Therapy.

Professionals were also informed via a recruitment post in different private healthcare Facebook groups. A total of 1,083 respondents completed the survey. Most respondents used KT for post-injury treatment (74 %), pain modulation (67 %), and neuro-sensory feedback (60 %). Most believed that KT stimulates skin mechano-receptors (77 %), improve local circulation (69 %), and modulates pain (60 %). Some respondents believed KT only created a placebo effect (40 %) and use it for such therapeutic purposes (58 %). Most used a standard uncut roll (67 %) in black (71 %) or beige (66 %). Most respondents did not use any specialty pre-cut tape (83 %), infused tape (99.54 %), or a topical analgesic with tape (65 %). The most common tape tension lengths used by respondents were 50 % tension (47 %) and 25 % (25 %) tension. Patient reported outcomes (80 %) were the most common clinical measures. Most respondents provided skin prep (64 %) and tape removal (77 %) instructions. Some did not provide any skin prep (36 %) or tape removal (23 %) instruction. The average recommended times to wear KT were 2 to 3 days (60%). The maximum times ranged from two to five days (81%). The authors concluded that this survey provided insight into how professionals use KT and highlighted the gap between research and practice. Moreover, these researchers stated that professionals also believe KT provides numerous positive therapeutic effects for clients; however, little is known regarding how the therapeutic effects might be produced with KT application. The KT conflicting results may be caused by 2 primary issues: tape manufacturing and study method differences. Future research addressing these 2 issues should be pursued to validate or refute the effectiveness of KT.

Intermittent Vertical Lumbar Traction

Vanti et al (2021) stated that only low-quality evidence is currently available to support the effectiveness of different traction modalities in the treatment of lumbar radiculopathy (LR). Yet, traction is still very commonly used in clinical practice. Some researchers have suggested that the subgroup of patients presenting signs and symptoms of nerve root compression and unresponsive to movements centralizing symptoms may benefit from lumbar traction. In a systematic review of RCTs, these investigators examined on the effects of vertical traction (VT) on pain and activity limitation in patients affected by LR. They searched the Cochrane Controlled Trials Register, PubMed, CINAHL, Scopus, ISI Web of Science and PEDro from their inception to March 31, 2019 to retrieve RCTs on adults with LR using VT to reduce pain and activity limitation. These researchers considered only studies reporting complete data on outcomes; 2 reviewers selected the studies, extracted the results, and carried out the quality assessment using the Risk of Bias and GRADE tools. A total of 3 studies met the inclusion criteria. Meta-analysis was not possible due to the heterogeneity of the included studies. These investigators found very-low quality evidence for a large effect of VT added to bedrest when compared to bedrest alone ($g = -1.01$; 95 % CI: -2.00 to -0.02). Similarly, VT added to medication may have a large effect on pain relief when compared to medication alone ($g = -1.13$; 95 % CI: -1.72 to -0.54, low quality evidence). Effects of VT added to PT on pain relief were very small when compared to PT without VT ($g = -0.14$; 95 % CI: -1.03 to 0.76, low quality evidence). All reported effects concerned short-term effect up to 3 months post-intervention. The authors concluded that VT may be an effective treatment only for reducing pain in LR at short-term; and may be preferred over passive treatments as bedrest and medications; however, VT did not demonstrate significant effects on activity limitation due to LR. These investigators stated that they had insufficient data to conclude that VT provided additional benefits when combined to or compared with PT. They stated that large, high-quality studies are needed to examine the

effectiveness of VT and identify the most effective delivering, the best treatment dosage, or the pain stage that could benefit more by this intervention.

The authors stated that the main drawbacks were related to the small number of included studies due to the very restricted population these investigators considered, as well as the small sample sizes of included studies. It did not allow a sensitivity analysis; however, these researchers tried to account for the risk of bias (RoB) found in the different studies with the GRADE method. Only 2 studies considered physical functioning as outcome measure, and only 1 study separately measured lumbar and sciatic pain. The overall summarizing of RoB was different from current standards, having also used the "moderate risk of bias" classification. The unreported standard deviations were derived from other similar studies: this may have led to an under-estimation or an over-estimation of the results. No study was rated as "high quality", and these researchers did not find any published protocols, making it difficult to evaluate reporting bias. They could not draw conclusions on what type of VT is better or which is the best patient's position. Due to the heterogeneity in treatment dosage, both in terms of time of application and days of treatment; therefore, suggestions on this topic were not forthcoming.

Johnson et al (2023) noted that lumbar intervertebral disc height loss has been associated with spinal height change (SHC) and LBP, including stenosis. Non-invasive methods to improve disc height loss require forms of lying down, which are uncondusive to computer work. Intermittent VT integrated with seated computer work may provide ergonomic alternatives for increasing SHC to promote LBP relief. These researchers introduced a safe VT prototype and dosage to induce and measure SHC. Prototype comfort and LBP ratings were exploratory secondary objectives. A total of 41 subjects were stadiometry-measured for pre- and post-intervention SHC from seated VT at 35 % body weight removed, supine lying (SL), and sitting at a computer (SIT) without VT. Pain ratings were recorded for those self-reporting LBP. VT prototype evaluations were compiled from a 3-question, 7-point Likert-style survey. SHC increased by 3.9 ± 3.4 mm in VT, 1.7 ± 3.4 mm in SIT, and 4.3 ± 3.1 mm in SL ($p < 0.000$). Post-hoc findings were significant between VT and SIT ($p < 0.000$), and SL and SIT ($p < 0.000$). VT and SL LBP ratings both decreased, but not SIT. The authors concluded that intermittent seated VT is a promising alternative for postural relief during seated computer work, producing SHC similar to lying down without compromising workflow.

Guidelines on back pain from the North American Spine Society (2020) state: "In patients with subacute or chronic low back pain, traction is not recommended to provide clinically significant improvements in pain or function. Grade of Recommendation: A."

Sustained Acoustic Medicine

Winkler et al (2021) stated that musculoskeletal injuries account for 10 million work-limited days per year and often result in both acute and/or chronic pain, and increased chances of re-injury or permanent disability. Conservative therapeutic options include various modalities, non-steroidal anti-inflammatory drugs (NSAIDs), and physical rehabilitation programs. Sustained acoustic medicine (SAM) is an emerging prescription home-use mechano-transductive device to stimulate cellular proliferation, increase micro-streaming and cavitation in-situ, and to increase tissue perfusion and permeability. These investigators examined the clinical evidence on SAM and measurable outcomes in the literature. They carried out a systematic literature review using PubMed, EBSCOhost, Academic Search Complete, Google Scholar and ClinicalTrials.gov to identify studies examining the effects of SAM on the musculoskeletal system of humans. Studies identified were selected based on inclusion criteria and scored on the Downs and Black checklist. Study design, clinical outcomes and primary findings were extracted from included studies for synthesis and meta-analysis statistics. A total of 372 subjects were included in 13 clinical research studies reviewed including 5 level-I, 4 level-II and 4 level-IV studies. A total of 67 subjects with neck and back myofascial pain and injury, 156 subjects with moderate-to-severe knee pain and radiographically confirmed knee osteoarthritis (OA; Kellgren-Lawrence grade II/III), and 149 subjects with generalized soft-tissue injury of the elbow, shoulder, back and ankle with limited function. Primary outcomes included daily change in pain intensity, change in Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), change in global rate of change (GRC), and functional outcome measures including dynamometry, grip strength, range-of-motion (ROM), and diathermic heating. The authors concluded that SAM treatment provided tissue heating and tissue recovery, improved patient function and reduction of pain. When patients failed to respond to PT, SAM proved to be a useful adjunct to facilitate healing and

return-to-work . As a non-invasive and non-narcotic treatment option with an excellent safety profile, SAM may be considered a good therapeutic option for practitioners.

The authors stated that although this systematic review focused on SAM for the treatment of musculoskeletal injuries, it is possible that other relevant studies using similar treatment parameters (3 MHz ultrasound [US] at 1.3 W) are available in the scientific literature to further aggregate and synthesize the clinical literature. This limitation was beyond the scope of this research but could be considered in a future analysis paying close attention to time, duration, dose delivered and regularity of US treatment. The literature search strategy found 13 relevant articles specific to SAM that were more than other past reviews on SAM ever, it was possible that some relevant studies were missed that were not available in English language or those in the grey literature that were emerging on this new therapeutic treatment. Furthermore, several of the outcome variables used in the studies differed in both measure, physical location on the body, condition being treated and control group that limited the scope of meta-analysis.

In a systematic review, Papalia et al (2023) examined the effectiveness of non-invasive procedures in relieving chronic pain due to failed back surgery syndrome (FBSS). Since patients who suffered from FBSS are often non-responders to analgesics, these investigators compared visual analog scale (VAS) for LBP and leg pain, Oswestry Disability Index (ODI), trial success rate, AEs and complications between conservative treatment groups and control groups. A total of 15 studies were included in this review. Spinal cord stimulation (SCS) was carried out in 11 trials; 4 studies examined the effectiveness of different epidural injections; 1 study examined repetitive transcranial magnetic stimulation (rTMS). All the studies reported back and leg pain relief following treatment with SCS, with a significant superiority in high-frequency (HFS) group, compared to low frequency (LFS) group. Moreover, disability decreased with each non-invasive treatment examined. Epidural injections of steroids and hyaluronidase have shown controversial results; AEs were described in 7 studies: lead migration, hardware-related events, infection and incisional pain were the most reported. Finally, trial success rate showed better outcomes for HFS. The authors concluded that this systematic review highlighted the effectiveness of conservative treatments in FBSS patients, with an improvement in pain scores and a decrease in disability index, especially after SCS with HFS. However, due to the lack of homogeneity among trials and population characteristics, further studies are needed to confirm the effectiveness of non-invasive interventions in patients affected by FBSS. Furthermore, sustained acoustic medicine is not mentioned as a therapeutic option.

Combined Capacitive Resistive Monopolar Radiofrequency and Proprioceptive Neuromuscular Facilitation

In a parallel, single-blind, single-center RCT, Jurak et al (2025) examined the effectiveness of combined capacitive resistive monopolar radio-frequency (CRMRF) and proprioceptive neuromuscular facilitation (PNF) training in the management of patients with chronic LBP (CLBP). Given the multi-factorial nature of CLBP, this study examined a multi-modal therapeutic approach integrating CRMRF, known for its thermal effects and ability to alleviate pain via improved cell metabolism and microcirculation, with PNF training, which enhances muscle strength, flexibility, as well as proprioception. This study was carried out in an outpatient clinical setting. Over the course of 4 months, a total of 62 patients, suffering from CLBP were randomly assigned to receive either the combined CRMRF and PNF treatment or PNF alone, with primary outcomes measured in terms of pain and functional disability using the VAS, and ODI. For secondary outcome of disability associated with pain, Quebec Pain Disability Scale (QPDS) and Roland-Morris Disability Questionnaire (RMDQ) were employed. The study's hypothesis was that the combined treatment would reduce pain and disability more effectively than PNF alone. Results indicated that the experimental group experienced greater improvements in pain and functional disability, surpassing the minimally clinically important difference (MCID) for the VAS, ODI, QPDS and RMDQ, suggesting the clinical relevance of the combined CRMRF and PNF approach. The authors concluded that the findings of this study were consistent with previous research highlighting the benefits of CRMRF in various musculoskeletal disorders and suggested that integrating CRMRF with PNF training would offer a promising non-invasive therapeutic option for patients with CLBP. These researchers stated that the results of this trial contributed to the growing evidence base supporting innovative, multi-modal treatment strategies for managing CLBP, with the potential to enhance patients' QOL.

The authors stated that this trial had several drawbacks. First, the absence of a

placebo group presented a significant limitation. Due to the inherent characteristics of CRMRF treatment, specifically its distinctive heating effect, it was not feasible to implement a convincing placebo condition that could replicate the sensation without delivering the actual therapeutic effect. This limitation was notable because the absence of a placebo control made it challenging to fully discount the placebo effect, which could potentially contribute to the observed benefits. The psychological impact of receiving a novel or technologically advanced treatment could influence patient-reported outcomes; therefore, the specific effects attributable solely to CRMRF, independent of placebo responses, remained uncertain. Second, the lack of mid-term and long-term follow-up evaluations in this trial restricted the understanding of the durability of treatment effects. Without these follow-up data, it was difficult to determine whether the improvements in pain and functional capacity observed post-treatment were sustained over time. This is a critical consideration for chronic conditions like CLBP, where the long-term management of symptoms is a key concern for both patients and clinicians. Future investigations should include placebo or sham-controlled conditions, when feasible, and incorporate follow-up evaluations to ascertain the longevity of treatment benefits; thus, providing a more comprehensive understanding of the therapeutic value of CRMRF combined with PNF exercises in the management of CLBP. Third, given fairly low sample size (total of 62 subjects) and the single-center design of this study, limitations were imposed on the ability to generalize these findings.

Matrix Rhythm Therapy

Ansari et al (2024) stated that CLBP is a widespread issue, and matrix rhythm therapy (MRT), a non-invasive therapy using low-frequency (LF) vibrations, is gaining interest for CLBP. However, evidence for its safety and effectiveness remains unclear. In a scoping review, these researchers examined the existing literature on using MRT to manage CLBP. A comprehensive search of Medical Literature Analysis and Retrieval System Online (Medline), PubMed, Web of Science, Directory of Open Access Journals (DOAJ), and Scopus was carried out for RCTs published through March 2024. The methodological quality was systematically assessed using the Physiotherapy Evidence Database Scale (PEDro). The initial search yielded a total of 47 studies. After screening the abstracts and full articles, 2 studies were included, with a mean score of 5/10 on the PEDro scale. A qualitative review was conducted of the 2 selected RCTs. Meta-analysis was not possible because of the small number of articles in the study. These researchers stated that while preliminary findings from the 2 studies suggested that MRT may be superior to Pilates and combined PT for pain and disability in CLBP, these findings require cautious interpretation because of limited research. The authors concluded that further high-quality studies, especially well-designed RCTs involving large numbers of participants, are needed to examine the effectiveness of MRT in managing patients with CLBP. These studies should evaluate the effect of MRT on pain levels, functional disability, as well as other subjective and objective parameters.

Unstable Footwear

Tedeschi et al (2024) noted that CLBP is a prevalent condition resulting in significant disability globally. Traditional treatments often have limited effectiveness, prompting research into alternative, non-invasive interventions. In a scoping review, these researchers examined the impact of unstable footwear on CLBP management, exploring its potential to reduce pain and disability. They analyzed RCTs focusing on adults with CLBP. Studies comparing unstable footwear to standard care or conventional footwear were included. Main outcomes measured were pain reduction, functional improvement, as well as biomechanical changes. Methodological quality was assessed using the PEDro scale. A total of 5 studies met the inclusion criteria, revealing a consistent trend towards pain reduction in subjects using unstable footwear. Specifically, intervention groups reported significant improvements in pain levels and, in some cases, disability scores compared to control groups. However, findings on functional performance and long-term effects were mixed, with no significant differences in certain studies. Methodological quality varied, with an estimated PEDro score range of 6 to 7 across the reviewed studies. The authors concluded that unstable footwear may offer a beneficial non-invasive intervention for patients with CLBP, contributing to pain reduction and possibly improving disability. Nonetheless, the evidence was tempered by methodological limitations and the need for further research. These investigators stated that future studies should focus on long-term outcomes, larger sample sizes, and clarifying the mechanisms behind the observed benefits. Clinicians should consider unstable footwear as part of a personalized treatment plan for CLBP, keeping abreast of new research to inform

their practice.

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